

Glycoin[®] natural

THE CELL ENERGIZER

ANTI-AGING BY THE MODEL OF NATURE

Glycoin[®] natural is a multifunctional anti-aging and cell-boosting active ingredient. It reactivates the power of youth in tired, stressed and mature skin:

- » stimulation of skin renewal, growth factors, ATP, SOD, CAT
- » instant hydration after one application by 49% (*in vivo*)
- » anti-aging & rejuvenation (*in vivo*)
- » microbiome support
- » blue light protection
- » brightening and lightening (*in vivo*)
- » 48 h tissue repair and wound healing (*ex vivo*)



Myrothamnus
plant during
dry season



Myrothamnus
plant during
rainy season



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PRODUCT DESCRIPTION

Glycoin® natural
THE CELL ENERGIZER

Glycoin® natural is a 100% natural multifunctional anti-aging and cell boosting active ingredient.

INCI: Glyceryl Glucoside (and) Aqua/Water

Appearance: clear, colorless liquid

It targets aged or stressed skin cells with sluggish cell functions and metabolism as well as dry, mature and sagging skin with loss of elasticity and skin thickness. **Glycoin® natural** is pure and natural Glyceryl Glucoside, biotechnologically manufactured by a patented sustainable process at bitop AG in Germany. It consists of the specific stereoisomer which is exclusively present as a stress-protection molecule in the desert resurrection plant *Myrothamnus flabellifolia* and blue-green algae ("Spirulina").^{1,2,3}



Myrothamnus
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Myrothamnus
plant during
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The small sugar derivative belongs to the group of extremolytes.⁴ Extremolytes are stress-protection molecules which protect extremophilic plants and microorganisms from their lethal habitats and allow them to thrive in extreme conditions like salt lakes, deserts, arctic ice, or hot springs. **Glycoin® natural** is the protection and re-activation substance of the resurrection plant *Myrothamnus flabellifolia*. This fascinating plant thrives in the deserts of Africa under extreme conditions like long-term dryness, high temperatures, and intense UV radiation. They have the unique ability to survive very long periods of total dehydration in which the leaves of the plant shrink and appear dead. But during rare moments of rainfall in the desert, a miracle of nature happens: the plant comes back to life and re-greens within a few hours.⁵ After the rain stops, the plant dries out again, until the next miracle of resurrection.

The stress-protection molecule **Glycoin® natural** enables the fascinating existence and resurrection behavior of the *Myrothamnus* plant.

During the desiccation of the plant, **Glycoin® natural** protects its structures from damage. As soon as water is available again, **Glycoin® natural** revitalizes the plant by boosting and stimulating its cell functions.⁶

REVITALIZATION AND BOOST OF SLOWED CELL FUNCTIONS

These fascinating properties of **Glycoin® natural** are transferable to our skin with outstanding results.

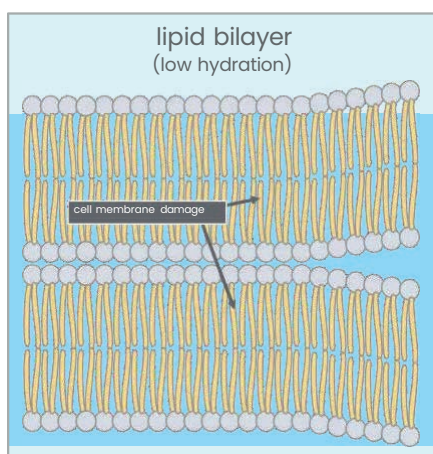
Glycoin® natural shows activity on aged and stressed skin cells with powerful rejuvenation properties. Old skin cells are revitalized and behave like young cells again. Sluggish cell functions and metabolic activity are boosted and stimulated.

Glycoin® natural is the perfect multifunctional anti-aging active ingredient for mature or tired and stressed skin – reactivating the power of youth by the model of nature.

Glycoin® natural is 100% natural (EcoCert, COSMOS), vegan, and worldwide applicable.

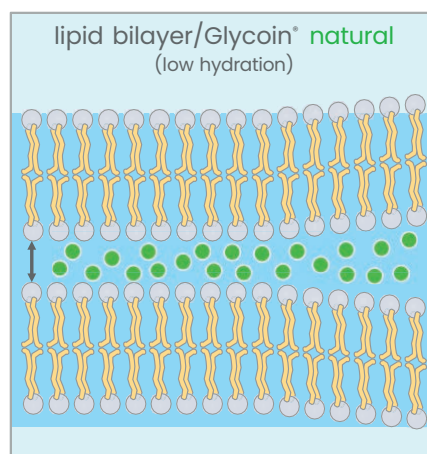
During total desiccation which can last for years, resurrection plants have to face the loss of all of their cellular water. For every other plant, this means total damage of cell structure and death. The stress protection molecule **Glycoin® natural** keeps the structures of the *Myrothamnus* plant alive and protects it from dying despite total desiccation. It protects the plant's cells by forming protection layers between the cell walls (see figure 2). Furthermore, due to the strong water retention capacity of **Glycoin® natural** (comparable to high molecular weight hyaluronic acid – see page 16), the hydration of the plant is ensured for as long as possible, even at low humidity of its habitat.

PROTECTION OF CELL MEMBRANE AT LOW HYDRATION STATUS



At low hydration status, less water is present between the membranes. This causes lateral compressive stress in the membrane plane and a transition from liquid to gel phase. This phase has a lower area per lipid molecule and a greater membrane thickness.

Figure 1: Membrane phase behavior at low hydration status without **Glycoin® natural**.



With **Glycoin® natural**, because of its osmotic and volumetric properties, the hydration will be higher, the separation between membranes will be larger, the lateral stress will be lower and the membrane remains in the fluid phase at low hydration status.

Figure 2: Membrane phase behavior at low hydration status with **Glycoin® natural**.

The main effect of so-called “osmolytes” like **Glycoin® natural** on membrane phase behavior is due to simple water interactions.⁷

Glycoin® natural has a positive effect on the osmotic pressure of the cell. Higher osmotic pressure results in a smaller suction (i.e., lower negative pressure) at any given negative water potential, meaning that less water is removed from between the cell membranes. The result is a larger average separation between structural components and the membranes itself (see figure 2).

Furthermore, the remaining **Glycoin® natural** molecules between the membrane bilayers have a volumetric effect during dehydration: the molecular volume of **Glycoin® natural** contributes to the physical separation of the bilayers at very low or zero water levels and thereby prevents their denaturation and damage.

Thus, the presence of small stress-protection molecules like **Glycoin® natural** diminishes stress and strain in the membranes at low or zero hydration levels.⁷

MEMBRANE STABILIZATION – CELL PROTECTION

In vitro studies were conducted to prove that **Glycoin® natural** is responsible for the stress-protection in the resurrection plant *Myrothamnus flabellifolia*.

For this, a *Myrothamnus* plant extract was tested for its cell protection efficacy and compared to **Glycoin® natural**.

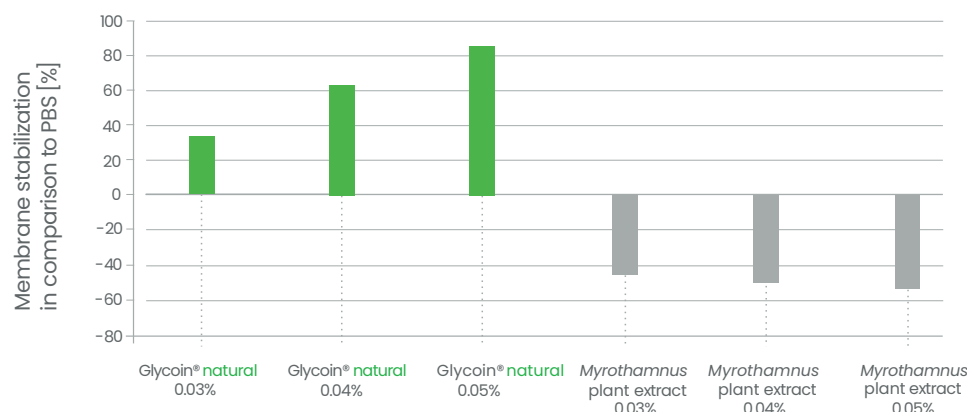


Figure 3: Membrane stabilization properties of Glycoin® natural compared to *Myrothamnus* plant extract.

Test protocol:

- *in vitro*, human epidermal keratinocytes
- pre-treatment with **Glycoin® natural** or an extract from *Myrothamnus* plant (0.03%, 0.04% and 0.05%)
- cells stressed with 0.001% SDS
- measured parameter: cell membrane stability

Results

Glycoin® natural protects the cell membranes of keratinocytes from damage.

The cell protection and stabilization properties of the *Myrothamnus* resurrection plant are directly linked to **Glycoin® natural**.

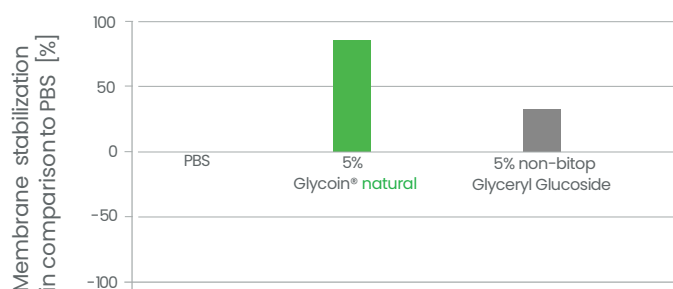


Figure 4: Membrane stabilization properties of Glycoin® natural compared to non-bitop Glyceryl Glucoside.

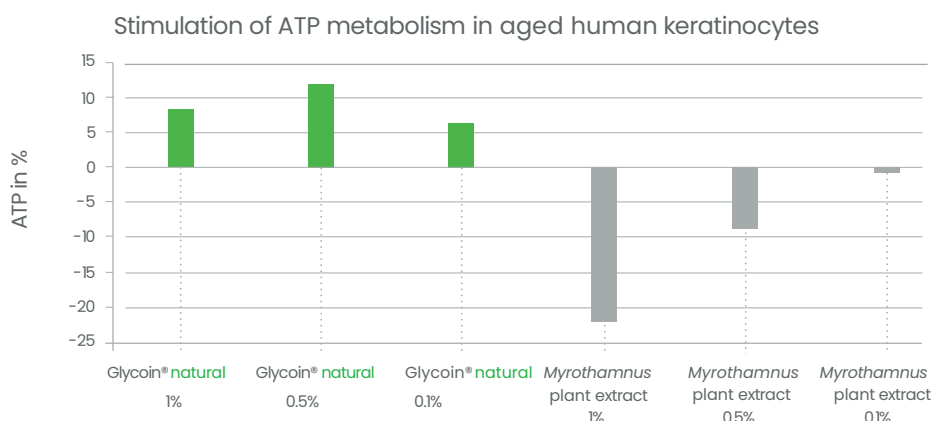
Test protocol:

- *in vitro*, human epidermal keratinocytes
- pre-treatment with 5% **Glycoin® natural** and non-bitop Glyceryl Glucoside
- cells stressed with 0.001% SDS for 10 min
- measured parameter: cell membrane stability

Results

Glycoin® natural, a 100% natural Glyceryl Glucoside which is present as a stress-protection substance in nature, demonstrates a much higher protection efficacy compared to non-bitop Glyceryl Glucoside.

STIMULATION AND BOOST OF SLUGGISH CELL FUNCTIONS



Test protocol:

- *in vitro*, human epidermal keratinocytes (HaCaT-cells); aging of cells for 2 weeks
- treatment with **Glycoin® natural** or an extract from *Myrothamnus* plant (1%, 0.5%, 0.1%); 6h incubation time
- measured parameter: ATP concentration in cells with luciferase assay

Figure 5: Change of ATP metabolism in aged human keratinocytes after pre-treatment with Glycoin® natural and a *Myrothamnus* plant extract at different concentrations.

Results

Compared to the plant extract, only **Glycoin® natural** stimulates the ATP production in aged skin cells.

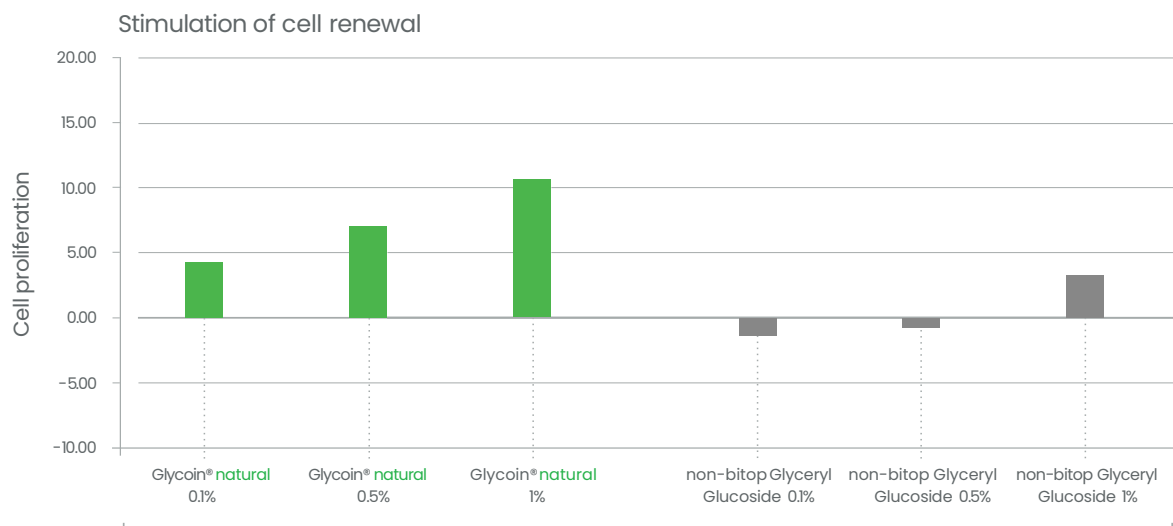


Figure 6: Stimulation of cell proliferation by Glycoin® natural and non-bitop Glyceryl Glucoside.

Results

Glycoin® natural is a 100% natural Glyceryl Glucoside which is present as a protection and cell boosting substance in nature. It shows a superior cell stimulation efficacy compared to non-bitop Glyceryl Glucoside.

STIMULATION AND ENERGIZING OF SLUGGISH CELL METABOLISM

During chronological aging (a result of the passing of time) and extrinsic skin aging (a result of external stress), skin cells show a clear reduction of metabolic activity and cell vitality. In skin cells, particularly of donors over the age of 40, metabolic activity declines as a function of age.

Several *in vitro* studies have shown the outstanding positive impact of **Glycoin® natural** on the metabolic activity of aged skin cells and cell vitality.

The cell vitality of human epidermal keratinocytes cultivated in the presence and absence of **Glycoin® natural** (1%) was measured.

Cell vitality was determined by the consumption of glucose in the keratinocytes. The more active the cells are, the more glucose they need for their metabolic activity. Cells cultivated in the presence of **Glycoin® natural** showed significantly higher metabolic activity than the untreated control.

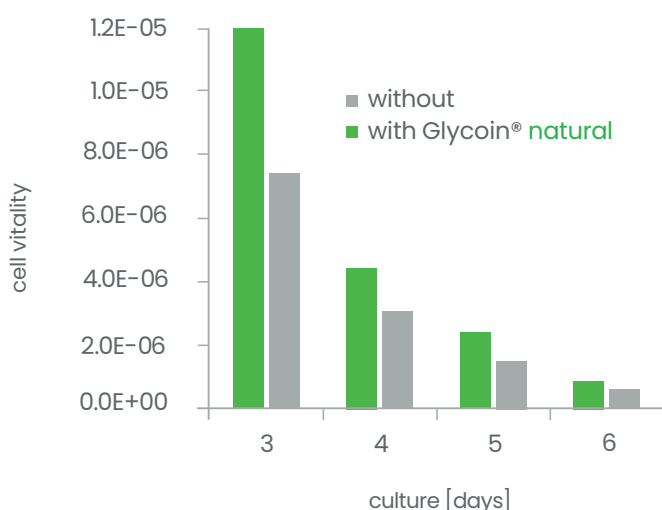


Figure 7: Vitality of cells during aging.

Results

Glycoin® natural increases the vitality and metabolic activity of aging skin cells by 170%.

STIMULATION OF CELL VITALITY – INCREASE OF ATP IN AGED CELLS

The skin is composed of many types of proliferating cells which have high energy demand. Adenosine triphosphate (ATP) – known as the bioenergy of life – is the vital source of energy for the metabolic activities of skin cells.

This includes all cellular activity, DNA replication, regeneration, and healing as well as collagen and elastin synthesis. Since the skin is the largest living organ of the body, it requires large amounts of ATP to thrive, regenerate and renew itself.

During aging (chronological and extrinsic), the amount of ATP in skin cells decreases constantly.⁹ A decline or dysfunction in ATP production impacts the skin's functions and appearance.⁹

The boost of ATP is a successful strategy to slow down visible skin aging and rejuvenate the skin. An increase of ATP leads to a stimulation of cell vitality and cell renewal, resulting in younger-looking skin.

Test protocol:

- *in vitro*, human keratinocytes from adult skin (HaCaT-cells); aging of cells for 2 weeks
- treatment with **Glycoin® natural**, glucose fructose, glycerin (0.5%, 0.75%, 0.1%); 6 h incubation time
- measured parameter: ATP concentration in cells with luciferase assay

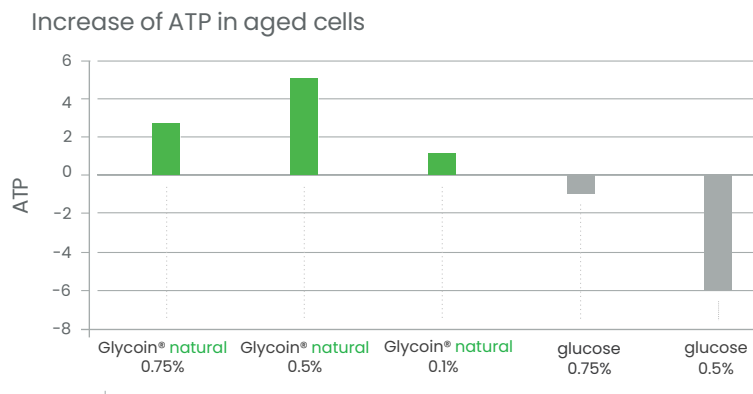


Figure 8: Increase of ATP concentration in aged human keratinocytes after pre-treatment with Glycoin® natural and glucose at different concentrations.

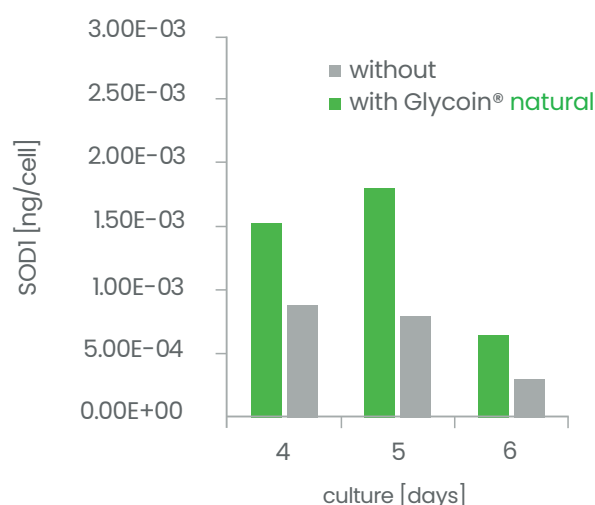
Results

Besides stimulating the cell metabolism, **Glycoin® natural** was also able to significantly increase the ATP concentration in aged skin cells.

Even at an advanced cell age stage when glucose, the gold standard for ATP stimulation, had no positive effect on the vitality, **Glycoin® natural** significantly increased the ATP concentration in these skin cells.

BOOST OF ANTIOXIDANTS IN AGING SKIN CELLS (SOD1)

An *in vitro* study conducted on human epidermal keratinocytes (HEKs) showed that the presence of 1% **Glycoin® natural** resulted in an increased production of the anti-oxidative skin protection enzyme "Superoxide Dismutase 1" (SOD1). With an advanced aging process of the cells, this effect even increased. **Glycoin® natural** treated cells produced 280% more SOD1 than untreated cells.



Superoxide Dismutase 1 (SOD1) is an important antioxidant naturally found in the human body. It is part of the skin's self-defense system. The enzyme reduces free radical damage in the skin, therefore preventing signs of aging. It supports wound healing, softens scar tissue, and protects against UV rays. Natural Superoxide Dismutase levels in the body drop as we age.⁸

Figure 9: Amount of SOD1 per cell during aging.

Results

Glycoin® natural increases the amount of SOD1 in aging skin cells by 280% and consequently strengthens the skin's own immune system. **Glycoin® natural** is an antioxidant booster.

INCREASE OF PROCOLLAGEN I PRODUCTION IN AGED CELLS

The stimulation of procollagen I synthesis was shown in an *in vitro* study, using the replicative aging model (Hayflick) on aged dermal fibroblasts. Aged fibroblasts were treated with **Glycoin® natural** in different concentrations (0.17%, 0.5% and 1.5%). **Glycoin® natural** showed a significant stimulation of procollagen I synthesis at all concentrations.

Results

Glycoin® natural significantly stimulates the procollagen I production of aged fibroblasts.

BLUE LIGHT PROTECTION

“Visible light” is part of the sunlight spectrum and is visible to the human eye (wavelength: 390–700 nm). It constitutes around 39% of the solar radiation and includes the high energy visible (HEV) light, better known as blue light (400–525 nm). In the past years, it was demonstrated that blue light irradiation induces a decrease in the expression of Aquaporin 3 (AQP3) in the epidermis and dermis.

AQP3 is the most abundant Aquaporin, also known as water and glycerin channel, in the epidermis. The role of AQP3 is not fully described as it is involved in different processes in the cell like wound healing processes.¹⁰ Furthermore, a reduction in AQP3 level has been associated with conditions like psoriasis, leukoplakia, diabetes mellitus, and aging.^{11,12,13,14,15,16}

Glycoin® natural is capable of preventing a decrease in the level of AQP3 in the cells induced by visible/blue light, thereby maintaining the hydration status of the skin.

In an *in vitro* study, NHEK were exposed to blue light (220 J/cm²) daily for three days to show the boosting effect on important cell functions of **Glycoin® natural**.

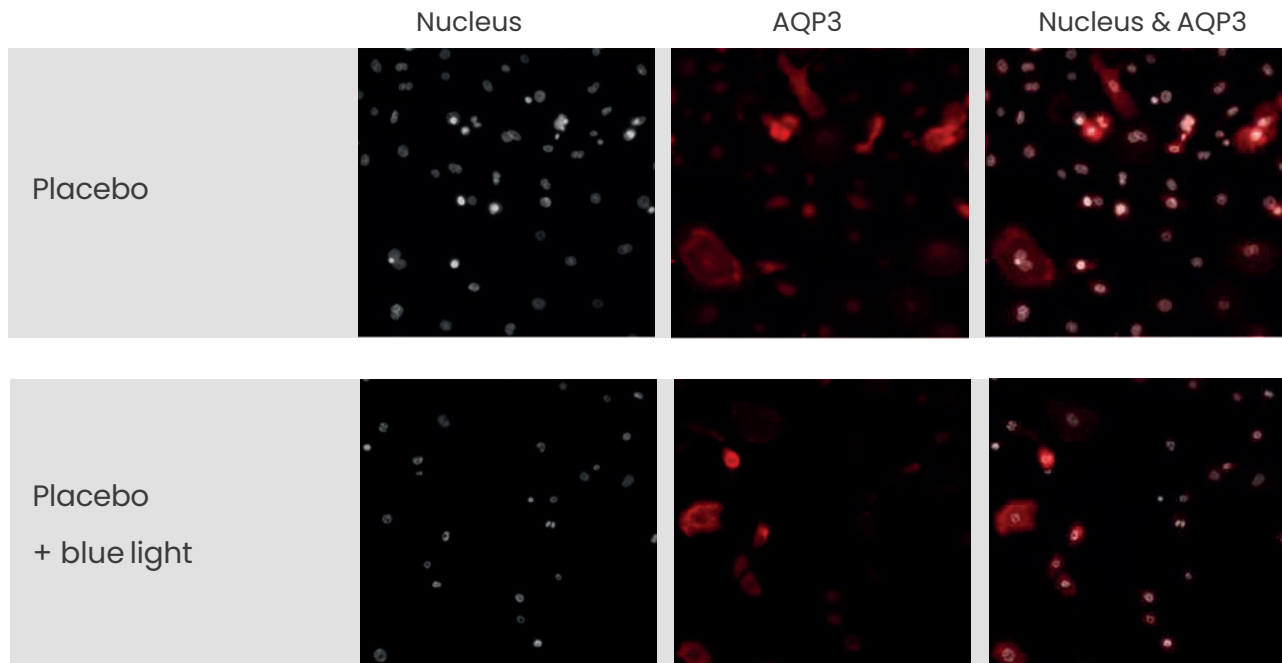


Study protocol

- normal human epidermal keratinocytes (NHEK, 23 years old, Caucasian woman)
- 0.5% **Glycoin® natural** in growth medium (Promocell) vs. placebo; incubation for 24 h before and throughout blue light exposure
- blue light exposure: 220 J/cm² (470 nm), 1 h 50 min daily for 3 days
- measured parameters: immunolabeling of Aquaporin 3 (AQP3)

AQP3 EXPRESSION (*IN VITRO*)

A – Placebo



B – Glycoin® natural

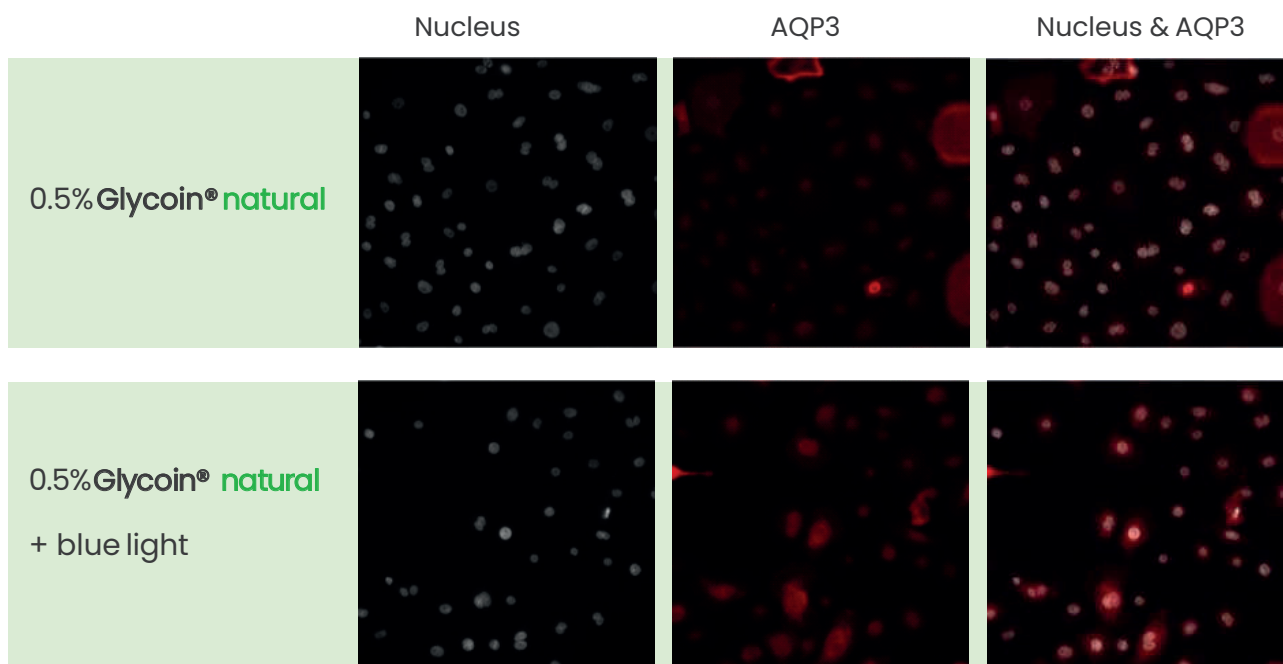


Figure 10: Fluorescence images of NHEK cells (AQP3 stained in red, nuclei stained in grey) (A) placebo before and after blue light irradiation (220 J/cm², 1 h 50 min daily for 3 days); (B) 0.5% Glycoin® natural before and after blue light irradiation (220 J/cm², 1 h 50 min daily for 3 days).

AQP3 EXPRESSION (*IN VITRO*)

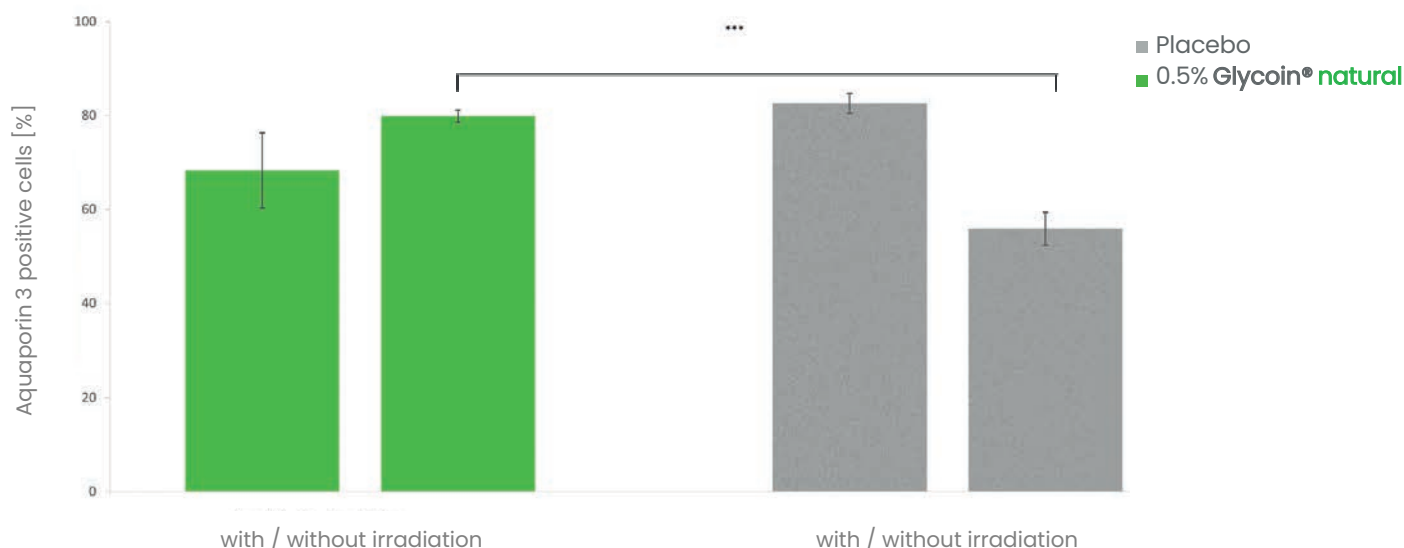


Figure 11: AQP3 positive NHEK cells [%] before and after blue light irradiation (220 J/cm², 1 h 50 min daily for 3 days) of placebo (grey) and 0.5% Glycoin® natural (green).

Results

After daily exposure (1 h 50 min) to blue light for three days, a significant decrease in AQP3 content in the NHEK cells was observed in the placebo control. The reduction in AQP3 may result in an imbalance in water and glycerol exchange.

Keratinocytes treated with 0.5% **Glycoin® natural** showed a significantly higher amount of AQP3 positive cells compared to the placebo by boosting the AQP3 expression during the blue light-induced stress. **Glycoin® natural** maintains the natural content of AQP3 in keratinocytes.

INFLUENCE ON THE HUMAN MICROBIOME



The human skin is home to millions of microorganisms, such as bacteria, viruses, or fungi that compose the individual skin microbiota. These microscopic organisms play an important role in maintaining the health of our skin by building a protective layer against invading pathogens.¹⁷ Furthermore, they are essential for the education of our immune system. A healthy microbiome helps the skin to stay in a good condition by stabilizing pH value and producing beneficial nutrients. Topical products can affect the microbiome positively by maintaining (“microbiome-friendly”) or even promoting the natural growth of microorganisms (prebiotic).

An *in vitro* study was carried out to evaluate the influence of **Glycoin® natural** on four bacterial strains that represent a large percentage of the human skin microbiome.

Study protocol

- mix of 4 bacterial strains (*Staphylococcus aureus* ATCC® 6538™, *Staphylococcus epidermidis* ATCC® 14990™, *Cutibacterium acnes* ATCC® 6919™, *Corynebacterium striatum* ATCC® 6940™)
- incubation in growth medium (EpiLife™) with 0.5% & 1% **Glycoin® natural**, control (placebo) for 48 h
- measured parameter: influence on the microbiome by measuring the growth of the bacterial strains; evaluation of bacterial viability by performing a XTT assay

PREBIOTIC EFFECT ON THE HUMAN MICROBIOME

Growth curve of bacterial mix

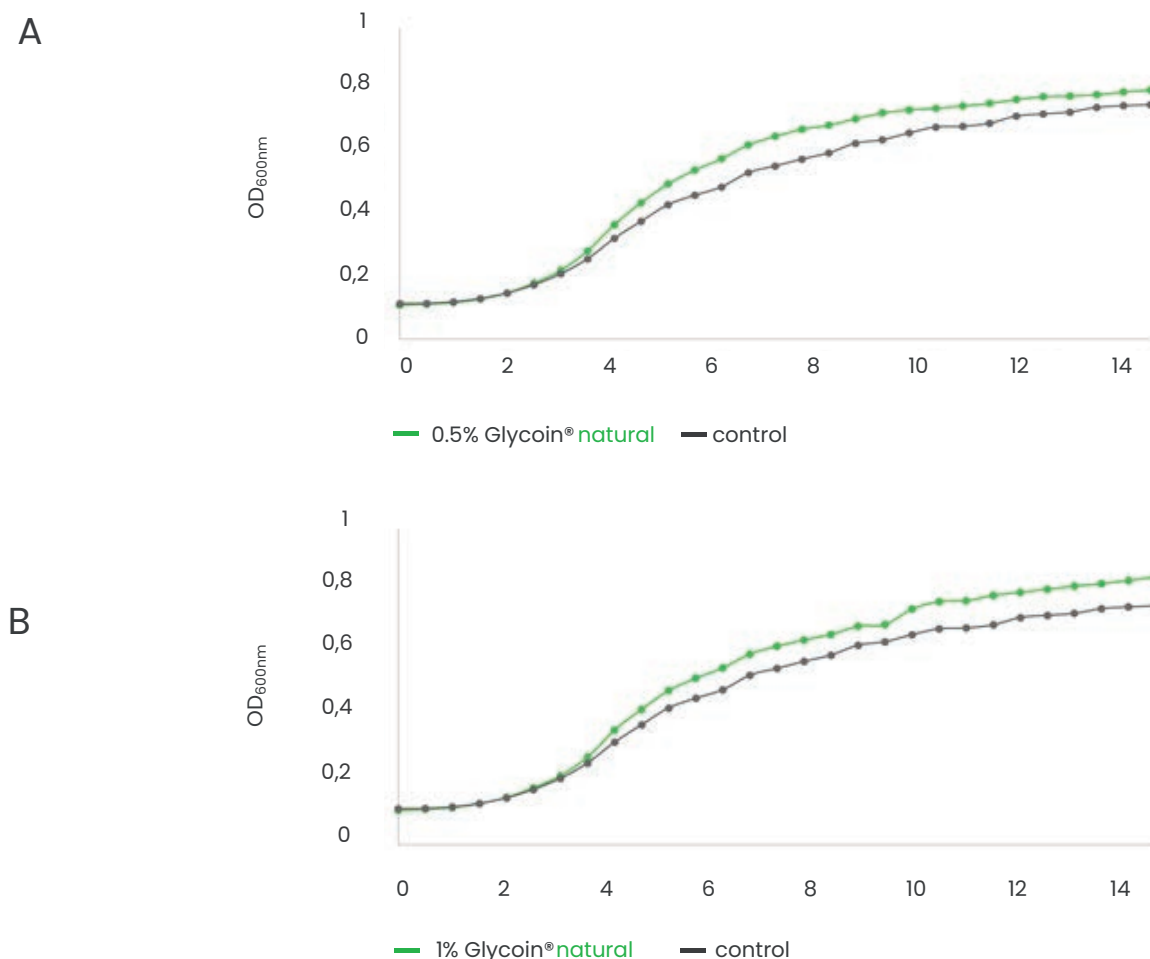


Figure 12: Growth curve of bacterial mix of (*Staphylococcus aureus* ATCC® 6538™, *Staphylococcus epidermidis* ATCC® 14990™, *Cutibacterium acnes* ATCC® 6919™, *Corynebacterium striatum* ATCC® 6940™) incubated with (A) 0.5% Glycoin® natural (green) and placebo (grey) (B) 1% Glycoin® natural (green) and placebo (grey); growth from t0 to t14h.

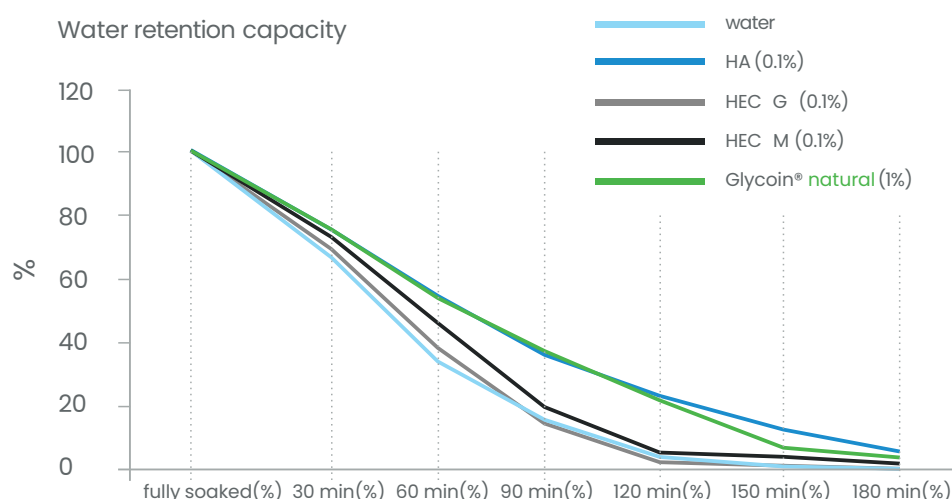
Results

0.5% & 1% **Glycoin® natural** enhanced bacterial growth over time and maintained the viability of the microorganisms after 48 h of incubation. 1% **Glycoin® natural** showed a higher effect compared to 0.5%.

Glycoin® natural is microbiome-friendly and additionally has got a prebiotic effect on the human microbiome by showing a growth-promoting activity.



WATER RETENTION CAPACITY – HYALURONIC ACID LIKE (LAB TEST)



Test protocol:

- Filter paper was weighed out and soaked in the test substances and again weighed out. Afterwards the papers were dried with interruptions to measure the weight of the drying filter papers.

Figure 13: Water retention capacities of HA (high molecular weight), HEC (Hydroxyethyl Cellulose G and M type), Glycoin® natural and H₂O.

Results

Glycoin® natural shows an almost identical water retention capacity like high molecular weight hyaluronic acid, but it is a much smaller molecule and has no tacky skin feeling.

ALL-DAY (8 h) MOISTURIZING (*IN VIVO*)

Test protocol:

- placebo-controlled *in vivo* study with 20 test persons, application on inner forearm
- single application of placebo and cream containing 1% and 0.5% **Glycoin® natural**
- measured parameter: skin hydration with Corneometer®

Table 1: Increase in skin hydration relative to initial conditions and to untreated in %.

increase of skin hydration compared to untreated	with 1% Glycoin® natural	with 0.5% Glycoin® natural
after 2 h	+ 49.5%	+ 49.5%
after 4 h	+ 34.5%	+ 32.2%
after 8 h	+ 24.7%	+ 21.3%

Results

Glycoin® natural hydrates the skin for 8 h after only one application. The skin moisture was increased by up to 49.5%. 100% of the test persons had a significant 8 h moisturization effect.

INCREASE IN SKIN THICKNESS AND DENSITY

Adult skin thins progressively at a rate that accelerates with age. The epidermis thins by up to 50% between the ages of 30 to 80. Changes in epidermal thickness are most pronounced in exposed areas such as face, neck, and décolleté as well as hands and forearms. Overall epidermal thickness decreases at about 6.4% per 10 years, decreasing faster in women than in men. Dermal thickness also is reduced with age. The loss of dermal collagen and elastin makes up most of the reduction in total skin thickness in elderly adults: for example, in postmenopausal women, a decrease in skin thickness of 1.13% per year parallels a 2% decrease per year in collagen content.⁹ The aim of the *in vivo* study was to show the positive effect of on the collagen and elastin production in the dermis and the resulting increase in dermal and epidermal thickness.

Test protocol:

- placebo-controlled, blinded, randomized *in vivo* study with 5 Caucasian test persons with mature skin
- assessment on inner forearm
- application of placebo and cream containing 1% **Glycoin® natural** twice-daily for 4 weeks
- measured parameter: epidermal and dermal density by ultrasonography (DermaScan C®)

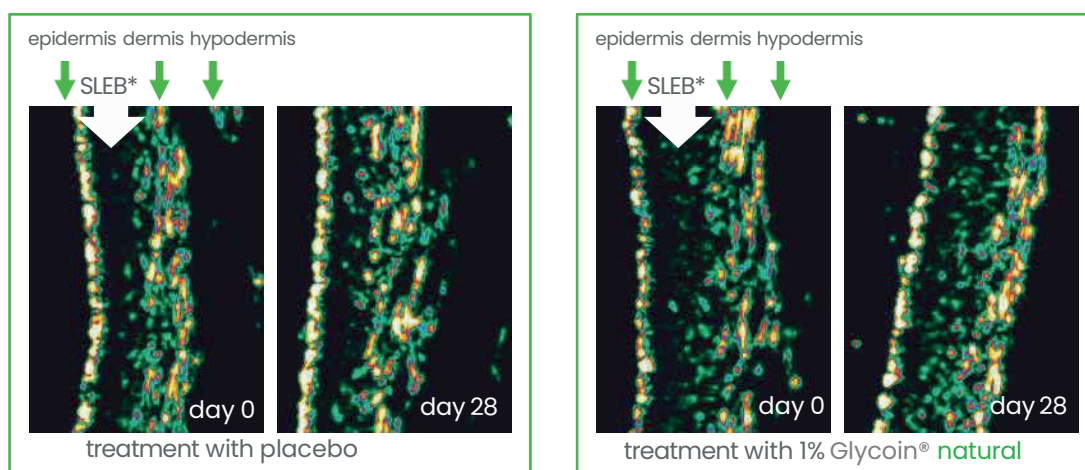


Figure 14: Ultrasound pictures of the skin showing the results of the epidermal and dermal density analysis. The reduction of darkpixels on a single picture equals the increase of epidermal and dermal thickness over time.

* subepidermal low echogenicity band of the skin

Results

Glycoin® natural improves the skin density of mature skin by up to 9.9% after four weeks compared to untreated skin.

100% of the test persons had a significant increase in skin thickness with **Glycoin® natural**.

INCREASE IN SKIN MOISTURE, ELASTICITY AND SMOOTHNESS

An *in vivo* study conducted on 20 female test persons, aged 45 and older, confirmed the revitalizing and stimulating efficacy of **Glycoin® natural** and demonstrated excellent and multifunctional anti-aging properties. Especially the facial elasticity (cheeks) was improved remarkably after four weeks of treatment with 1% **Glycoin® natural**.

Test protocol:

- placebo-controlled, blinded *in vivo* study with 20 Caucasian test persons, 4 weeks
- assessment on cheeks (elasticity), lower leg (moisture and smoothness)
- application of placebo and cream containing 1% **Glycoin® natural** twice-daily for 4 weeks
- measured parameter: skin moisture (Corneometer®), elasticity (Cutometer®) and smoothness (PRIMOS®)

Table 2: *In vivo* study results after 4 week application.

Skin properties	with 1% Glycoin® natural, after 28 days	with 1% Glycoin® natural, compared to placebo after 28 days
Skin moisture (Corneometer®)	+ 24.9%	+ 23.3%
Skin elasticity (Cutometer®)	+ 8.3%	+ 93.3%
Skin smoothness (PRIMOS)	+ 27.0%	+ 61.8%

Results

Glycoin® natural increases the skin moisture by 24%, skin smoothness by 61%, and the skin elasticity by up to 93%.

The increase of skin elasticity was measured in the cheek area, showing a firming efficacy of **Glycoin® natural**.

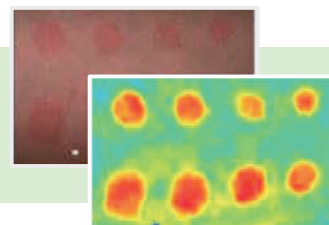
REDUCTION OF SUNBURN AND SKIN REDNESS (*IN VIVO*)

The back of 20 male and female test persons with skin phototype II and III was irradiated with UV light. The irradiated areas were treated 3 times (7 h, 24 h, and 32 h after irradiation) with placebo cream, 2% **Glycoin® natural** cream, and cream containing 0.5% Asiaticoside. After 48 h the skin redness (with chromameter) and skin microcirculation (with TiVi®) on the irradiated areas were measured.

76% of the test persons treated with **Glycoin® natural** showed a soothing effect on sunburn compared to 41% treated with Asiaticoside and 35% treated with placebo cream.

Results

Glycoin® natural soothes sunburn after only three times of application.



EFFICACY – *IN VIVO*

Glycoin[®] natural
THE CELL ENERGIZER

BRIGHTENING & LIGHTENING EFFICACY (*IN VIVO*)



BRIGHTENING & LIGHTENING EFFICACY (*IN VIVO*)

Due to environmental factors such as sun exposure, air pollution, and exposure to blue light, the skin develops pigmentation – unwanted freckles and sun spots increasingly arise.

The aim of the in vivo study was to show the brightening efficacy of **Glycoin® natural**.

Test protocol:

- *in vivo* study with 22 female Asian test persons (44–60 years old) with photo-type III and IV
- subjects presenting at least one pigment spot with a diameter ≥ 3 mm on the face
- application of cream containing 1% **Glycoin® natural** on face twice-daily for 84 days.
- on day 0, 28, 56 and 84: measurement with Spectrocolorimeter®.
- on day 84 subjective evaluation.
- measured parameter: depigmenting effect, reduction of the different pigmentations, subjective questionnaires

Results: Depigmenting effect

Table 3: Parameters for depigmenting effect

Parameter	Change in % after 56 days	Change in % after 84 days
Skin lightness	+ 2%	+ 2%
Skin pigmentation	- 36%	- 38%

Results

Glycoin® natural significantly increased the skin lightness by up to 2%. The skin pigmentation decreased significantly by up to 38% after 84 days of application.

Results: Reduction of pigmentation

Characterized by a significant decrease of the variation between spot and normal skin

Table 4: Parameters for the reduction of different pigmentations.

Parameter	Change in % after 56 days	Change in % after 84 days
variation between spot and normal skin	-11%	-21%

Results

The variation between pigmented skin and normal skin significantly decreased by up to 21%.

Glycoin® natural is able to reduce pigmentation.

Results of the subjective questionnaires

Table 5: Product efficacy after 84 days of use.

Statement	% of positive answers
The size of spots is smaller	96%
The color of spots is attenuated (less dark)	96%
The number of spots is decreased	87%
The skin is lighter	91%
The skin is radiant	91%
The skin appears smoother	95%
The product leaves a uniform complexion	95%
The product leaves a softer skin	100%
The product leaves a hydrated skin	100%

Results

The subjective questionnaire shows a satisfaction by the large majority of the subjects for the properties and efficacy of **Glycoin® natural**.

TISSUE REPAIR AND STIMULATION OF GROWTH FACTORS (*EX VIVO*)

Wound healing is an important adaptive response against infection of the skin and is needed for the maintenance of homeostasis. The healing process begins immediately after the injury and occurs in three phases: inflammation and migration, proliferation, and remodeling and maturation.

The re-epithelialization involves migration, and proliferation of keratinocytes and fibroblasts, in addition to the production of cytokines and growth factors that affect other cells. Wound healing trials (ex vivo skin model and cDNA array) revealed impressive tissue repair properties of **Glycoin® natural**. Injured areas recovered and healed faster than the placebo group. After only 48 h significant wound-healing efficacy was shown for **Glycoin® natural** treated areas.

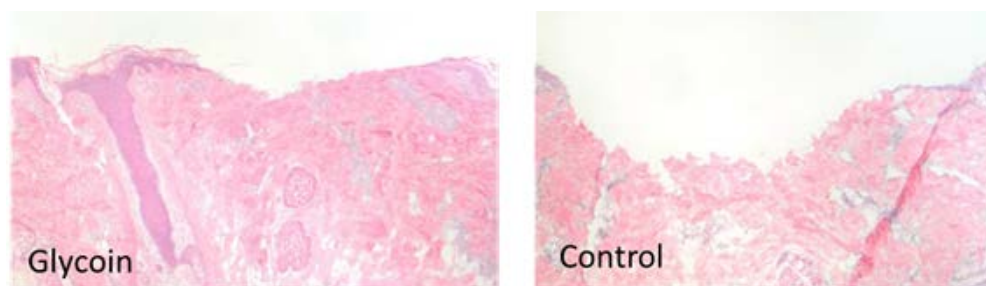
One of the reasons for this efficacy is the stimulation of cell renewal by **Glycoin® natural**. As shown before, **Glycoin® natural** stimulates cell proliferation and has a powerful boosting effect on cell metabolism.

Furthermore, **Glycoin® natural** induces the expression of growth factors involved in the wound healing process: FGF7 and TGF-beta 1, initiating the proliferation of new dermal cells. Additionally, the expression of so-called ROS scavenger molecules (CAT, SOD2, TRDX) is enhanced by **Glycoin® natural**. The expression of these molecules is essential for proper tissue repair.¹⁸

Test protocol:

- ex vivo skin model; surface was mechanically reduced by 3 mm (loss of epidermis and dermis)
- immediately after wound making, wound spot was covered with 5 µl PBS with 2.4% **Glycoin® natural** and without (placebo)
- incubation for 48 h
- hematoxylin/eosin staining of skin model for evaluation

Results



treatment with Glycoin® natural

treatment with placebo (control)

Figure 15: Wound healing demonstration on ex vivo skin model after 48 h.

Results

Glycoin® natural demonstrated excellent tissue repair efficacy and stimulation of growth factors (e.g. FGF7, TGF-beta1) after only 48 h.

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safe
GMO-free
100% natural
halal
China compliant
non allergenic
Ecocert
COSMOS
vegan
preservative free

Product description

Applications

- » skin care (face, body)
- » after-sun care
- » oral care
- » baby care
- » wound care
- » color cosmetics

suitable for extremely sensitive and allergic skin

Formulation guidelines

- » level of use 0.5–2%
- » no incompatibility with other ingredients
- » can be added directly to aqueous phase prior to emulsification (hot or cold)

bitop's **Glycoin® natural** is enantiopure and 100% natural Glycerol Glucoside. It exclusively consists of the stereoisomer which is present as a stress-protection and revitalization substance in the resurrection plant "*Myrothamnus flabellifolia*" and blue-green algae ("Spirulina").

Technical data

INCI	Glycerol Glucoside (and) Aqua/Water
CAS number	22160-26-5
Description	100% natural small molecule, (enantiopure) substance, sugar derivate
Chemical name	2-O- α -D-Glucopyranosylglycerine
Molecular formula	C ₉ H ₁₈ O ₈
Molecular weight	254.24 g/mol
Concentration	≥ 50% Glycerol Glucoside dissolved in water
Appearance	clear, colorless liquid, slightly characteristic odor
pH value	4.0 – 7.5
Stability	pH 3–11, light and oxygen stable, temperature stable (≤ 85 °C)
Solubility	excellent solubility in water
Shelf life	3 years
Storage conditions	5 °C – 25 °C or freeze
Level of use	0.5–2%
Pack sizes	1 kg, 5 kg, 20 kg
Production process	enzymatic reaction / patented biocatalytic process
Origin	• revitalization and protection substance of the desert plant <i>Myrothamnus flabellifolia</i> • stress protection substance of salt tolerant blue-green algae ("Spirulina")

bitop
Extremolytes for life

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